

3. Researchers have created an artificial eye that models the constriction and dilation of the pupil. The change in size of the pupil in response to changing light levels is an example of a reflex arc.

Image 3.1A shows a diagram of the eye and **Image 3.1B** shows parts from the artificial eye model.

Image 3.1A

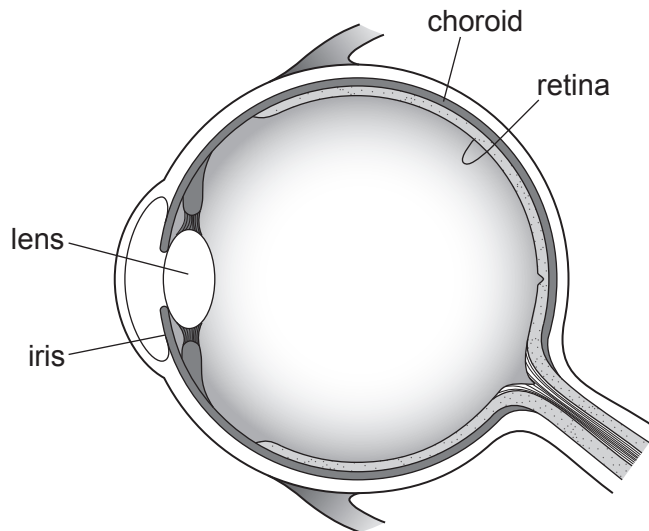
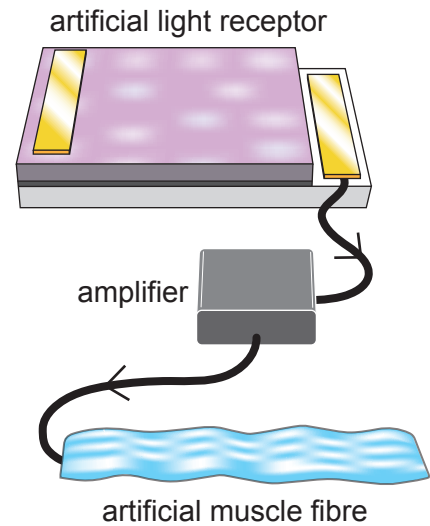


Image 3.1B



- (a) Draw an arrow labelled **A** on **Image 3.1A** to show the sclera. [1]
- (b) State **two** properties of a reflex arc. [1]
1.
2.
- (c) State the name of the part of the eye represented by the artificial muscle fibre in **Image 3.1B**. [1]
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- (d) Describe how the diameter of the pupil changes in response to bright light. State the advantage of this change to the eye. [2]
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